Cambridge (CIE) IGCSE Computer Science

Your notes

Cyber Security

Contents

Cyber Security Threats

Your notes

Forms of cyber security threat

- Computers face a variety of forms of attack and they can cause a large number of issues for a network and computers
- The main threats posed are:
  - **Brute-force attacks**
  - **Data interception & theft**
  - **DDos attack**
  - **Hacking**
  - **Malware**
  - **Pharming**
  - **Phishing**
  - **Social engineering**

Brute Force Attack

What is a brute-force attack?

- A brute force attack works by an attacker repeatedly trying **multiple combinations** of a user's password to **try and gain unauthorised access** to their accounts or devices
- An example of this attack would be an attacker finding out the length of a PIN code, for example, 4-digits
- They would then **try each possible combination** until the pin was cracked, for example
  - **0000**
  - **0001**
  - **0002**
- A second form of this attack, commonly used for passwords is a **dictionary attack**
- This method tries **popular words or phrases** for passwords to guess the password as quickly as possible
- Popular words and phrases such as '**password**', '**1234**' and '**qwerty**' will be checked extremely quickly.

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Data interception

What is data interception & theft?

- Data interception and theft is when thieves or hackers can compromise **usernames** and **passwords** as well as other sensitive data
- This is done by using devices such as a **packet sniffer**
- A packet sniffer will be able to **collect the data** that is being transferred on a network
- A thief can use this data to gain **unauthorised access** to websites, companies and more

DDoS Attack

What is a DDoS attack?

- A **Distributed Denial of Service Attack** (DDoS attac**k**) is a **large scale**, **coordinated** attack designed to **slow down a server** to the point of it becoming **unusable**
- A server is continually **flooded with requests** from **multiple distributed devices** preventing genuine users from accessing or using a service
- A DDoS attack uses computers as '**bots**', the bots act as **automated tools** under the attackers control, making it difficult to **trace back** to the original source
- A DDoS attack can result in companies **losing money** and not being able to carry out their daily duties
- A DDoS attack can cause **damage to a company's reputation**

Hacking

What is hacking?

- Hacking is the process of **identifying and exploiting weaknesses** in a computer system or network to **gain unauthorised access**
- Access can be for various **malicious purposes**, such as stealing data, installing malware, or disrupting operations
- Hackers **seek out opportunities** that make this possible, this includes:
  - **Unpatched software**
  - **Out-of-date anti-malware**

Malware

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What is malware?

- Malware (**mal**icious soft**ware**) is the term used for any software that has been created with malicious intent to cause harm to a computer system
- Examples of issues caused by malware include
  - Files being **deleted**, **corrupted** or **encrypted**
  - Internet connection becoming **slow** or **unusable**
  - Computer **crashing** or **shutting down**
- There are various types of malware and each has slightly different issues which they cause

Malware	What it Does

Virus	▪ Contains code that will **replicate** and cause **unwanted and unexpected events** to occur
	▪ Examples of issues a user may experience are
	  ▪ **Corrupt** files
	  ▪ **Delete** data
	  ▪ **Prevent** applications from running correctly

Worms	▪ Very similar to viruses, main difference being that they **spread to other drives** and **computers** on the network
	▪ Worms can infect other computers from
	  ▪ **Infected websites**
	  ▪ **Instant message services**
	  ▪ **Email**
	  ▪ **Network connection**

Trojan	▪ Sometimes called a **Trojan Horse** ▪ Trojans **disguise** themselves as **legitimate** software but contain **malicious** code in the background
Spyware	▪ Allow a person to **spy** on the users' **activities** on their devices

	▪ Embedded into other software such as games or programs that have been downloaded from **illegitimate sources** ▪ Can **record** your screen, log your **keystrokes** to gain access to **passwords** and more
Adware	▪ Displays **adverts** to the user ▪ Users have **little or no control** over the **frequency** or **type** of ads ▪ Can **redirect clicks** to unsafe sites that contain spyware
Ransomware	▪ Locks your computer or device and **encrypts** your documents and other important files ▪ A **demand is made for money** to receive the password that will allow the user to decrypt the files ▪ **No guarantee** paying the ransom will result in the user getting their data back

Pharming

What is pharming?

- Pharming is typing a website address into a browser and it being **redirected to a 'fake' website** in order to trick a user into typing in sensitive information such as passwords
- An attacker attempts to **alter DNS settings**, the directory of websites and their matching IP addresses that is used to access websites on the internet or change a users browser settings
- A user **clicks a link** which **downloads malware**
- The user **types in a web address** which is then **redirected** to the fake website

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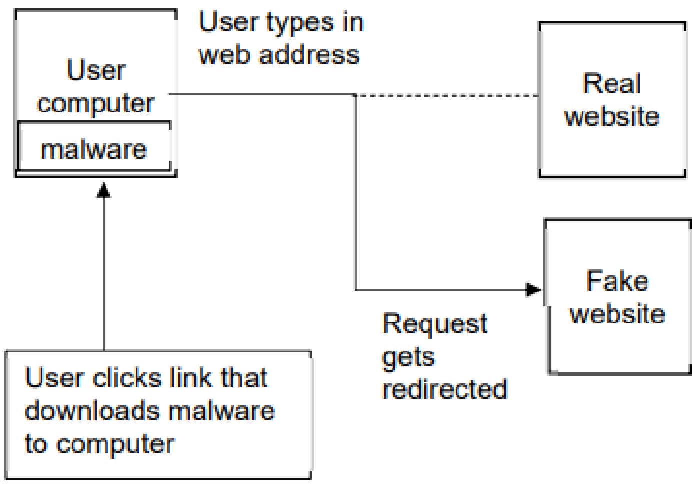

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How can you protect against it?

- To protect against the threat of pharming:
  - **Keep anti-malware software up to date**
  - **Check URLs regularly**
  - **Make sure the padlock icon is visible**

Phishing

What is phishing?

- Phishing is the process of sending fraudulent emails/SMS to a large number of people, claiming to be from a **reputable company** or trusted source
- Phishing is an attempt to try and **gain access** to your details, often by coaxing the user to click on a login button/link

Social Engineering

What is social engineering?

- Social engineering is **exploiting weaknesses** in a computer system by **targeting the people** that **use** or **have access** to them
- There are many forms of social engineering, some examples include
  - **Fraudulent phone calls**: pretending to be someone else to gain access to their account or their details

- **Pretexting:** A scammer will send a fake text message, pretending to be from the government or human resources of a company, this scam is used to trick an individual into giving out confidential data
- People are seen as **the weak point** in a system because human errors can lead to significant issues, some of which include:
  - **Not locking doors to computer/server rooms**
  - **Not logging their device when they're not using it**
  - **Sharing passwords**
  - **Not encrypting data**
  - **Not keeping operating systems or anti-malware software up to date**

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Worked Example

A company is concerned about a distributed denial of service (DDoS) attack.

(i) Describe what is meant by a DDoS attack.

[4]

(ii) Suggest one security device that can be used to help prevent a DDoS attack. **[1]**

Answers

(i) Any four from:

- multiple computers are used as bots
- designed to deny people access to a website
- a large number / numerous requests are sent (to a server) ...
- ... all at the same time
- the server is unable to respond / struggles to respond to all the requests
- the server fails / times out as a result.

(ii)

- firewall **OR** proxy server

Keeping Data Safe

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Access Levels

What are access levels?

- Access levels ensure users of a network **can access what they need to access** and do not have access to information/resources they shouldn't
- Users can have **designated roles** on a network
- Access levels can be set based on a user's **role, responsibility, or clearance level**
  - **Full access –** this allows the user to open, create, edit & delete files
  - **Read-only access –** this only allows the user to open files without editing or deleting
  - **No access –** this hides the file from the user
- Some examples of different levels of access to a school network could include:
  - **Administrators:** *Unrestricted* - Can access all areas of the network
  - **Teaching Staff:** *Partially restricted* - Can access all student data but cannot access other staff members' data
  - **Students:** *Restricted* - Can only access their own data and files
- Users and groups of users can be given **specific file permissions**

Anti Malware

What is anti-malware software?

- Anti-malware software is a term used to describe a combination of different software to prevent computers from being susceptible to **viruses** and other **malicious software**
- The different software anti-malware includes are
  - **Anti-virus**
  - **Anti-spam**
  - **Anti-spyware**

How does anti-malware work?

- Anti-malware **scans** through **email** attachments, **websites** and downloaded **files** to search for issues

- Anti-malware software has a list of known malware **signatures to block** immediately if they try to access your device in any way
- Anti-malware will also perform **checks for updates** to ensure the database of known issues is up to date

Authentication

What is authentication?

- Authentication is the process of **ensuring that a system is secure** by asking the user to **complete tasks to prove they are an authorised** user of the system
- Authentication is done because **bots can submit data in online forms**
- Authentication can be done in several ways, these include
  - **Usernames and passwords**
  - **Multi-factor authentication**
  - **CAPTCHA** - see example below

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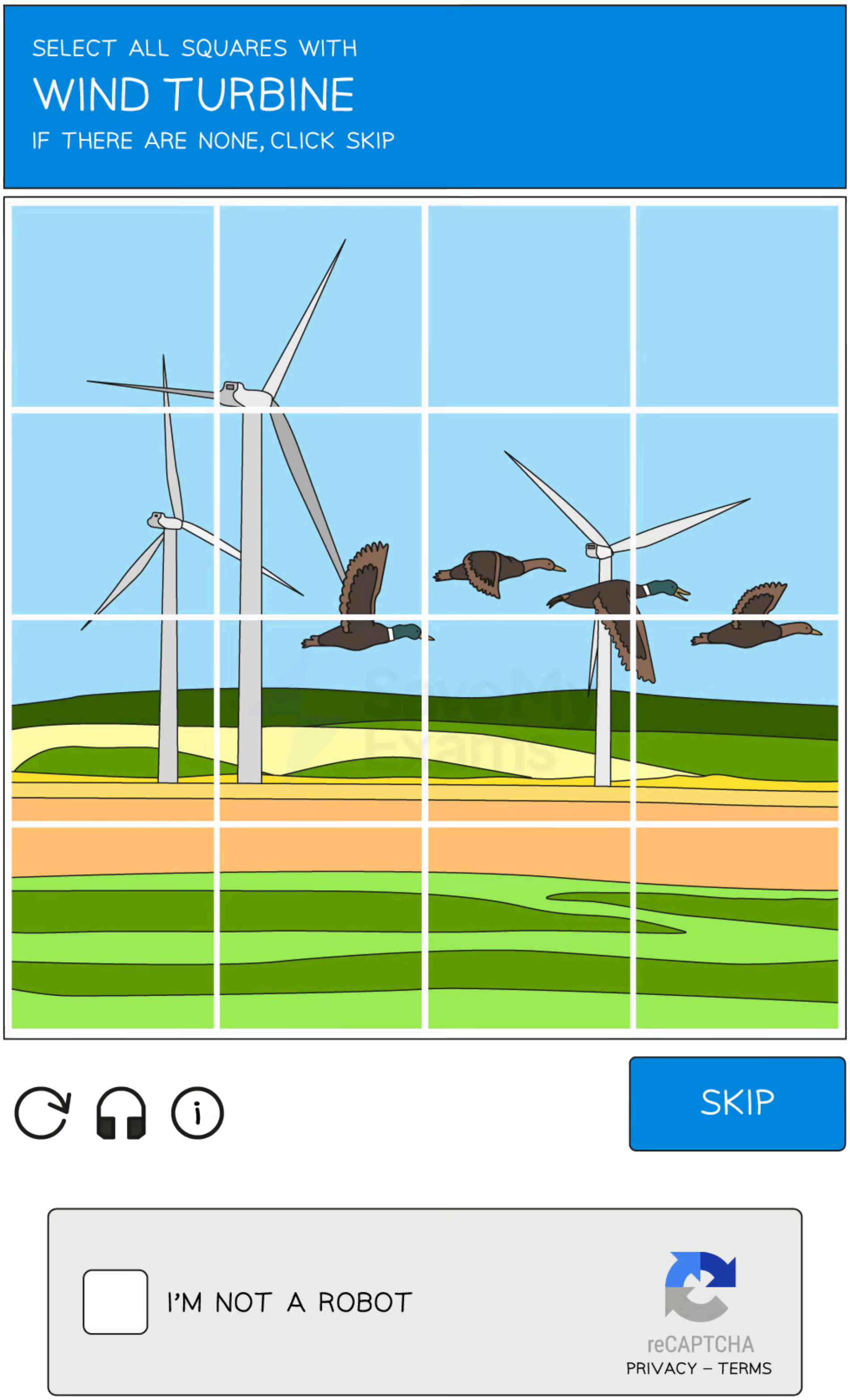
SELECT ALL SQUARES WITH
WIND TURBINE
IF THERE ARE NONE, CLICK SKIP
SKIP
I'M NOT A ROBOT
reCAPTCHA
PRIVACY – TERMS

Biometrics

- Biometrics use **biological data** for authentication by identifying **unique physical characteristics** of a human such as **fingerprints**, **facial recognition**, or **iris scans**
- Biometric authentication is **more secure** than using passwords as:
  - A biometric password **cannot be guessed**
  - It is very **difficult to fake** a biometric password
  - A biometric password **cannot be recorded** by spyware
  - A perpetrator **cannot shoulder surf** to see a biometric password

Automating Software Updates

What are automatic software updates?

- Automatic software updates take away the need for a user to remember to keep software updated and **reduce the risk of software flaws/vulnerabilities** being targeted in **out of date software**
- Automatic updates **ensure fast deployment** of updates as they release

Communication

What is communication?

- One way of protecting data is by **monitoring digital communication** to check for errors in the **spelling** and **grammar** or **tone** of the communication
- Phishing scams often involve communication with users, monitoring it can be effective as:
  - **Rushed** - emails and texts pretending to be from a reputable company are focused on quantity rather than quality and often contain basic spelling and grammar errors
  - **Urgency** - emails using a tone that creates panic or makes a user feel rushed is often a sign that something is suspicious
  - **Professionalism** - emails from reputable companies should have flawless spelling and grammar

URL

How to check a URL?

- Checking the **URL attached to a link** is another way to prevent phishing attacks

- Hackers often use fake URLs to trick users into visiting fraudulent websites
  - e.g. **http://amaz.on.co.uk/** rather than **http://amazon.co.uk/**
- If you are unsure, always check the website URL **before clicking any links** contained in an email

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Firewalls

What is a firewall?

- A firewall **monitors incoming and outgoing network traffic** and uses a set of rules to determine which traffic to allow
- A firewall prevents **unwanted traffic** from entering a network by filtering requests to ensure they are **legitimate**
- It can be both **hardware** and **software** and they are often used together to provide stronger security to a network
  - **Hardware firewalls** will protect the whole network and prevent unauthorised traffic
  - **Software firewalls** will protect the individual devices on the network, monitoring the data going to and from each computer

What form of attack would this prevent?

- Hackers
- Malware
- Unauthorised access to a network

Privacy Settings

What are privacy settings?

- Privacy settings are used to **control the amount of personal information** that is shared online
- They are an important measure to prevent identity theft and other forms of online fraud
- Users should **regularly review** their privacy settings and adjust them as needed

Proxy Servers

What is a proxy server?

- A proxy-server is used to **hide a user's IP address** and **location**, making it more difficult for **hackers to track them**

- They **act as a firewall** and can also be used to **filter web traffic** by setting criteria for traffic
- Malicious content **is blocked** and a warning message can be sent to the user
- Proxy-servers are a useful security measure for protecting against external security threats as it can direct traffic away from the server

SSL

What is SSL?

- Secure Socket Layer (**SSL**) is a **security protocol** which is used to **encrypt data** transmitted **over the internet**
- This helps to prevent **eavesdropping** and other forms of **interception**
- SSL is widely used to protect **online transactions**, such as those involving credit card information or other sensitive data
- It works by sending a **digital certificate** to the user's browser
- This contains the **public key** which can be used for **authentication**
- Once the certificate is authenticated, the transaction will begin

Worked Example

(i)) Identify a security solution that could be used to protect a computer from a computer virus, hacking and spyware.

Each security solution must be different

Threat	Security solution
Computer virus	
Hacking	
Spyware	

[3]

(ii) Describe how each security solution you identified in (i) will help protect the computer.

[6]

Answers

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(i)

Threat	Security solution
Computer virus	Anti-malware/virus (software) Firewall
Hacking	Firewall Passwords Biometrics Two-step verification
Spyware	Anti-malware/virus (software) Two-step verification Firewall

(ii) **Two** marks for each description

- **Anti-malware/virus (software)**
  - Scans the computer system (for viruses)
  - Has a record of known viruses
  - Removes/quarantines any viruses that are found
  - Checks data before it is downloaded
  - ... and stops download if virus found/warns user may contain virus
- **Anti-malware/spyware (software)**
  - Scans the computer for spyware
  - Removes/quarantines any spyware that is found
  - Can prevent spyware being downloaded
- **Firewall**
  - Monitors traffic coming into and out of the computer system
  - Checks that the traffic meets any criteria/rules set
  - Blocks any traffic that does not meet the criteria/rules set // set blacklist/whitelist
- **Passwords**
  - Making a password stronger // by example
  - Changing it regularly
  - Lock out after set number of attempts // stops brute force attacks // makes it more difficult to guess
- **Biometrics**
  - Data needed to enter is unique to individual
  - ... therefore it is very difficult to replicate

- Lock out after set number of attempts
- **Two-step verification**
  - Extra data is sent to device, pre-set by user
  - … making it more difficult for hacker to obtain it
  - Data has to be entered into the same system
  - … so if attempted from a remote location, it will not be accepted

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